

Project / Module: Portfolio

Date: 16/11/2025

Goals for the documented session

- Complete Express MongoDB + Mongoose exercise

Tasks Completed

- Completed Express MongoDB + Mongoose exercise

Use of Gen AI

- Prompt: How can I read a CSV and save the info into a DB?
- Output:

```
fs.createReadStream("people.csv")
  .pipe(csv())
  .on("data", async (row) => {
    try {
      const person = new Person({
        name: row.name,
        email: row.email,
        age: parseInt(row.age),
      });
      await person.save();
      console.log("Guardado:", row.name);
    } catch (err) {
      console.error("Error al guardar:", err);
    }
  })
  .on("end", () => {
```

```
console.log("CSV procesado completamente.");
```

```
});
```

- Edits: After I was sure that the code worked correctly, I erased console.log because it was for debugging. I adapted the code to the correct names and added the team schema.
- Reflection: AI helped me to understand the syntax for reading a csv and inserting the data into a DB, but I needed to validate every part of the code to ensure that it works for my specific structure and models. AI can be useful for guidance, but it is necessary to understand the code and test by myself to fully learn something.

- Output: How to convert the day separately?

```
const [day, month, year] = str.split("/").map(Number);
```

```
new Date(year, month - 1, day);
```

- Edits: I added this part to try before the schema to cast the date and then added it to the schema.
- Reflection: I didn't know that JavaScript recognized months from 0 to 11. I already knew how to apply this and how does it work but I couldn't remember the syntax.
- Prompt: ¿En qué momento se guarda en la colección? ¿Tengo mi schema drivers y teams y creé un modelo drivers y teams, leo el csv y se guarda en la colección drivers, pero cuando se guardara en la colección teams?
- Output:

```
for (const row of csvRows) {
```

```
  // Guardar driver
```

```
  await Driver.create({
```

```
    number: row.number,
```

```
    code: row.code,
```

```
    forename: row.forename,
```

```
    surname: row.surname,
```

```

    dob: new Date(row.dob), // asegurarte que es Date
    nationality: row.nationality,
    url: row.url,
    current_team: row.current_team
  });

```

```

// Guardar equipo si no existe

```

```

const existingTeam = await Team.findOne({ name: row.current_team });
if (!existingTeam) {
  await Team.create({ name: row.current_team });
}
}

```

- Edits: I adapted this code to mine because the structure wasn't what I wanted. I only used this to understand how to add it to a different model and how to verify that the team wasn't in the model before. I created a new Team object, related it to the driver's info and used save instead of create.
- Reflection: AI helped me to understand the general idea, but I needed to understand the code well to adapt it to my code.
- Prompt: I can't remember the first prompt but I told chat that I was having problems with accessing to the row.number value, it helped me to understand what was the problem and I found out that the key of number wasn't the one that it seems, surely because it was the first thing written in the csv, so I had to clean it.
- Output:

```

const cleanRow = {};
for (const key in row) {
  cleanRow[key.trim().replace(/^uFEFF/, "")] = row[key]; // quita espacios y BOM
}
const num = Number(cleanRow.number);

```

console.log(num).

- Edits: Inserted this part of the code into the part in my code where the file is read and added the variable num to the driver's object.
- Reflection: AI is a very helpful tool to debug and understand problems from the very root because I told it to tell me how to print the value from row.number to find out where the problem was. I discovered that row.number was undefined, and it was a very strange thing because in the csv the key said "number". After a lot of tests, chat told me to print to see the real hex value of the key and that's when I discovered that it was a BOM problem, but I didn't even know what it was before this so without AI's help, I wouldn't be able to solve the problem. I spent three days straight trying to solve this error.
- Prompt: "Cómo programo este elemento de bootstrap?"
- Output:

```
<div class="form-check form-switch">
```

```
  <input class="form-check-input" type="checkbox" id="toggleContent">
```

```
  <label class="form-check-label" for="toggleContent">Mostrar  
  contenido</label>
```

```
</div>
```

```
<div id="content" style="display:none;">
```

```
  <p>Contenido visible cuando el switch está activado.</p>
```

```
</div>
```

```
<script>
```

```
  const toggle = document.getElementById("toggleContent");
```

```
  const content = document.getElementById("content");
```

```
  toggle.addEventListener("change", () => {
```

```
    content.style.display = toggle.checked ? "block" : "none";
```

```
});
```

```
</script>
```

- Edits: Instead of using the ternary operator I used an if-else condition because it helped me to visualize better the flow of the code. I edited the name of the variables to adapt it to my code. I added a condition to change the label of the switch depending on its output.
- Reflection: It is very helpful to have a base when you don't know how to start writing the code to modify an element. From that base you can understand how it works and, with your knowledge, do whatever you want.
- Prompt: "My teams are being duplicated if there are two drivers in the same team in the CSV"

- Output:

```
let team = await Team.findOneAndUpdate(  
  { name: row.current_team },  
  { $setOnInsert: { id: null, nationality: null, url: null } },  
  { upsert: true, new: true }  
);
```

- Edits: I replaced this part for a condition and .save().
- Reflection: My old code wasn't working because await verified and made the teams almost at the same time so it didn't recognized that there were already one team with the same name. This code MongoDB makes one action at a time.
- Prompts: Pasted all the code in index.ejs and said "¿puedes hacer la interfaz más bonita?"
- Output: Changed the design with bootstrap. Added cards, divided everything into divs, used bootstrap classes, used the grid system. Created a CSS file with hovers, colors, fonts, margin, border radius, etc.
- Edits: Changed the background color and changed the submit button. I overwrite in the CSS .btn.btn-secondary and changed from the predefined blue color to a yellow.
- Reflection: Using AI for improving the design helps to save time, but it is important to understand the code so that if you want to make further changes you can know what to edit.

What I Learned

- `Driver.save()` vs `Driver.insertMany()`. It is more efficient to use `insertMany` and insert a lot of documents at the same time with a `forEach` when having a lot of rows in the csv. However, since the csv in the exercise has only twenty something rows, it is not a problem if I add them row by row.
- Schema: The structure for the documents. Models: There is one for drivers and one for teams. It interacts with the collections in the code. DB: In f1 is both the info of drivers and teams. Collections: The set of documents in Mongo.
- Although the csv could show that the columns name has some form, it can have another one, so when you try to access the value it sends undefined. The csv can have BOM issues and to save the variable correctly you must convert the keys in the csv before.
- BOM issue: When there is a hidden character at the very beginning of a file called Byte Order Mark (`\uFEFF`).
- Lean returns an POJO object which is better for reading the variable in EJS.
- `.findOneAndUpdate()` Search for a data in the BD and updates its values.

Challenges & Bugs)

- ❌ Problems with the type of value that the schema expects.
- ✅ The number wasn't casting correctly so I added a validation to manage those errors. Also, the date wasn't casting correctly so I took every number, saved it into separated variables and manually inserted it into a variable of type Date.
- ❌ Duplication of teams despite the condition.
- ✅ Change the condition, `team.save()` by `Team.findOneAndUpdate`.
- ❌ The driver's numbers aren't being saved correctly.
- ✅ Clean the row before casting the value to eliminate `\uFEFF` at the beginning of the file.

Resources Used

<https://sites.google.com/up.edu.mx/mongodb/home/mongodb-atlas>

Next Steps

- The last four exercises from the portfolio

Personal Reflection

(e.g.) “Today I felt more confident debugging server-side issues. Using Gen AI helped me move faster, but I made sure to understand the code before using it.”

Date: 17/11/2025

Goals for the documented session

- Complete BasicReact exercise

Tasks Completed

- Everything, but there is a minor I still must solve. When you click on like of one movie, all the buttons update.

Use of Gen AI

- Prompt: I'd like to add a scroll when the user clicks on More.
- Output:
- `useEffect(() => {`
- `if (selectedMovie) {`
- `const el = document.getElementById("character-section");`
- `el?.scrollIntoView({ behavior: "smooth" });`
- `}`
- `}, [selectedMovie]);`
- Edits: I integrated this code into mine before `return ()`. I imported `useEffect` and added an id to the corresponding div.
- Reflection: AI helped me to understand the basis to add a scroll, but it is important to understand the code so that you can integrate correctly into your code.
- Prompt: Help me create a function that saves the info from a form.
- Output:

```
const handleCommentSubmit = (e) => {
```

```
e.preventDefault(); // Evita que la página se recargue
```

```
const name = e.target.name.value.trim();
```

```
const comment = e.target.comment.value.trim();
```

```
if (!name || !comment) return; // No guardar si está vacío
```

```
setComments((prev) => {
```

```
  const movieComments = prev[selectedMovie.episode] || [];
```

```
  return {
```

```
    ...prev,
```

```
    [selectedMovie.episode]: [...movieComments, { name, comment }]
```

```
  };
```

```
});
```

```
e.target.reset(); // Limpia el formulario
```

```
};
```

- Edits: I integrated that part of the code into mine. I modified the code that AI gave me. Instead of using the id I used the e to read the fields from the forms.
const formData = new FormData(e.target); const name = formData.get("name");
I created the variable that saves the comments, implemented the call to the function into the html and added the onSubmit to the forms tag to save the info into the comments variable.
- Reflection: Reflection: AI helped me to understand the basis to save the comments with the form, but it is important to understand the code so that you can integrated correctly into your code. There are many ways in which you can do the same thing, you jus have to learn when to use what.

What I Learned

- `setTimeout`: It allows you to set a waiting time before an action occurs.
- `useRef()`: Sets a ref to an element in the DOM.

Challenges & Bugs

- ❌ The scroll wasn't working.
- ✅ Added `setTimeout` so that the page tried to scroll after setting the card into the DOM.
- ❌ After adding the hover feature, the more feature stop working.
- ✅ The problem was that I was missing a `#` in href.

Next Steps

- The last four exercises from the portfolio

Personal Reflection

(e.g.) "Today I felt more confident debugging server-side issues. Using Gen AI helped me move faster, but I made sure to understand the code before using it."